

Федеральное государственное автономное образовательное
учреждение высшего образования

Университет ИТМО

Дисциплина: Администрирование систем и сетей

Лабораторная работа 6

Работу выполнили студенты:

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Желаемая оценка: 3

Преподаватель:

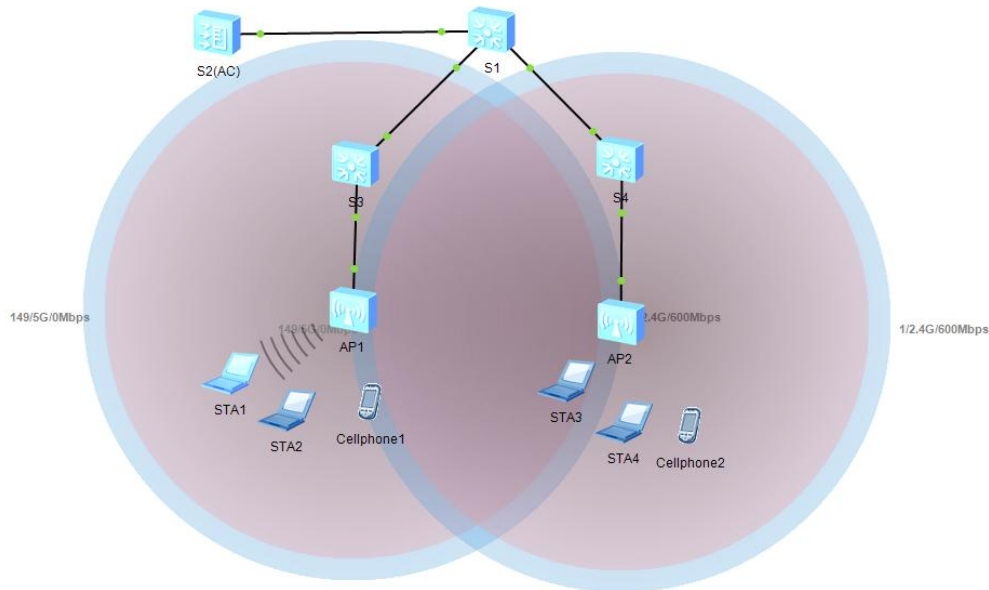
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Санкт-Петербург

Создание WLAN

Топология



Конфигурирование

Настройка параметров проводной сети

Настройка VLAN:

```
[S1]vlan batch 100 101
Info: This operation may take a few seconds. Please wait for a moment...done.
[S1]interface g0/0/13
[S1-GigabitEthernet0/0/13]port link-type trunk
[S1-GigabitEthernet0/0/13]port trunk allow-pass vlan 100 101
[S1-GigabitEthernet0/0/13]quit
[S1]interface g0/0/14
[S1-GigabitEthernet0/0/14]port link-type trunk
[S1-GigabitEthernet0/0/14]port trunk allow-pass vlan 100 101
[S1-GigabitEthernet0/0/14]quit
[S1]interface g0/0/10
[S1-GigabitEthernet0/0/10]port link-type trunk
[S1-GigabitEthernet0/0/10]port trunk allow-pass vlan 100 101
[S1-GigabitEthernet0/0/10]quit
[AC]vlan batch 100 101
```

Info: This operation may take a few seconds. Please wait for a moment...done.

[AC]interface g0/0/10

[AC-GigabitEthernet0/0/10]port link-type trunk

[AC-GigabitEthernet0/0/10]port trunk allow-pass vlan 100 101

[AC-GigabitEthernet0/0/10]quit

[S3]vlan batch 100 101

Info: This operation may take a few seconds. Please wait for a moment...done.

[S3]interface g0/0/1

[S3-GigabitEthernet0/0/1]port link-type trunk

[S3-GigabitEthernet0/0/1]port trunk allow-pass vlan 100 101

[S3-GigabitEthernet0/0/1]quit

[S3]interface g0/0/4

[S3-GigabitEthernet0/0/4]port link-type trunk

[S3-GigabitEthernet0/0/4]port trunk pvid vlan 100

[S3-GigabitEthernet0/0/4]port trunk allow-pass vlan 100 101

[S3-GigabitEthernet0/0/4]quit

[S4]vlan batch 100 101

Info: This operation may take a few seconds. Please wait for a moment...done.

[S4]interface g0/0/1

[S4-GigabitEthernet0/0/1]port link-type trunk

[S4-GigabitEthernet0/0/1]port trunk allow-pass vlan 100 to 101

[S4-GigabitEthernet0/0/1]quit

[S4]interface g0/0/4

[S4-GigabitEthernet0/0/4]port link-type trunk

[S4-GigabitEthernet0/0/4]port trunk pvid vlan 100

[S4-GigabitEthernet0/0/4]port trunk allow-pass vlan 100 to 101

[S4-GigabitEthernet0/0/4]quit

Настройка IP-адресов интерфейсов:

[S1]interface Vlanif 101

[S1-Vlanif101]ip addr 192.168.100.0 24

[S1-Vlanif101]quit

[S1]interface LoopBack 0

[S1-LoopBack0] ip address 10.0.1.1 32

[S1-LoopBack0]quit

[AC]interface Vlanif 100

[AC-Vlanif100]ip addr 192.168.100.254 24

Настройка DHCP:

[S1]dhcp enable

Info: The operation may take a few seconds. Please wait for a moment.done.

[S1]ip pool sta

Info:It's successful to create an IP address pool.

[S1-ip-pool-sta]network 192.168.101.0 mask 24

[S1-ip-pool-sta]gateway-list 192.168.101.254

[S1-ip-pool-sta]quit

[S1]interface Vlanif 101

[S1-Vlanif101]dhcp select global

[S1-Vlanif101]quit

[AC]dhcp enable

Info: The operation may take a few seconds. Please wait for a moment.done.

[AC]ip pool ap

Info: It is successful to create an IP address pool.

[AC-ip-pool-ap]network 192.168.100.254 mask 24

[AC-ip-pool-ap]gateway-list 192.168.100.254

[AC-ip-pool-ap]quit

[AC]interface Vlanif 100

[AC-Vlanif100]dhcp select global

[AC-Vlanif100]quit

Настройка параметров точек доступа для выхода в сеть

Создание группы AP ap-group1:

[AC]wlan

[AC-wlan-view]ap-group name ap-group1

Info: This operation may take a few seconds. Please wait for a moment.done.

[AC-wlan-ap-group-ap-group1]quit

Создание профиля регулирующего домена и настройка кода страны AC в профиле:

[AC]wlan

[AC-wlan-view]regulatory-domain-profile name default

[AC-wlan-regulate-domain-default]country-code on

Error: The country code is invalid.

Установка привязки профиля регулирующего домена к группе AP:

[AC]wlan

[AC-wlan-view]ap-group name ap-group1

[AC-wlan-ap-group-ap-group1]regulatory-domain-profile default

Warning: Modifying the country code will clear channel, power and antenna gain configurations of the radio and reset the AP. Continue?[Y/N]:Y

```
[AC-wlan-ap-group-ap-group1]quit
```

Указание интерфейса-источника на AC для установления туннелей CAPWAP:

```
[AC]capwap source interface Vlanif 100
```

Импорт точки доступа в AC и добавление их в группу AP с именем ap-group1:

```
[AC]wlan
[AC-wlan-view]ap auth-mode mac-auth
[AC-wlan-view]ap-id 0 ap-mac 00e0-fc08-2340
[AC-wlan-ap-0]ap-name ap1
[AC-wlan-ap-0]ap-group ap-group1
Warning: This operation may cause AP reset. If the country code changes, it will
clear channel, power and antenna gain configurations of the radio, Whether to continue?
[Y/N]:Y

[AC-wlan-ap-0]quit

[AC-wlan-view]ap-id 1 ap-mac 00e0-fceb-6ab0
[AC-wlan-ap-1]ap-name ap2
[AC-wlan-ap-1]ap-group ap-group1
Warning: This operation may cause AP reset. If the country code changes, it will
clear channel, power and antenna gain configurations of the radio, Whether to continue?
[Y/N]:Y

[AC-wlan-ap-1]quit
```

Вывод на экран информации о текущей AP:

```
[AC]wlan
[AC]dis ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor : normal      [2]
-----
-----
ID  MAC          Name Group   IP        Type        State STA Upti
me
-----
-----
0   00e0-fc08-2340 ap1  ap-group1  192.168.100.27 AP2050DN    nor  0
4S
1   00e0-fceb-6ab0 ap2  ap-group1  192.168.100.71 AP2050DN    nor  0
2S
-----
-----
```

Total: 2

Настройка параметров сервисов WLAN

Создание профиля безопасности HCIA-WLAN и настройка политики безопасности:

```
[AC-wlan-view]security-profile name HCIA-WLAN
[AC-wlan-sec-prof-HCIA-WLAN]security wpa-wpa2 psk pass-phrase HCIA-Datacom aes
[AC-wlan-sec-prof-HCIA-WLAN]quit
```

Создание профиля SSID HCIA-WLAN и задание имени SSID HCIA-WLAN:

```
[AC-wlan-view]ssid profile name HCIA-WLAN
[AC-wlan-ssid-prof-HCIA-WLAN]ssid HCIA-WLAN Info: This operation may take a
few seconds, please wait.done.
[AC-wlan-ssid-prof-HCIA-WLAN]quit
```

Создание профиля VAP HCIA-WLAN, настройка режима передачи данных и сервисного VLAN и применение профиля безопасности и профиля SSID к профилю VAP:

```
[AC-wlan-view]vap-profile name HCIA-WLAN
[AC-wlan-vap-prof-HCIA-WLAN]forward-mode direct-forward
[AC-wlan-vap-prof-HCIA-WLAN]service-vlan vlan-id 101
Info: This operation may take a few seconds, please wait.done. [AC-
wlan-vap-prof-HCIA-WLAN]security-profile HCIA-WLAN
Info: This operation may take a few seconds, please wait.done. [AC-
wlan-vap-prof-HCIA-WLAN]ssid-profile HCIA-WLAN
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-vap-prof-HCIA-WLAN]quit
```

Установка привязки профиля VAP к группе AP и применение конфигурации профиля VAP HCIA-WLAN к радиомодулю 0 и радиомодулю 1 точек доступа в группе AP:

```
[AC-wlan-view]ap-group name ap-group1
```

```
[AC-wlan-ap-group-ap-group1]vap-profile HCIA-WLAN wlan 1 radio all  
Info: This operation may take a few seconds, please wait...done.
```

Проверка

Перед проверкой нужно подключиться к WLAN!!!

Пароль: HCIA-Datacom

The screenshot shows the STA1 configuration window with the following details:

- MAC Address:** 54-89-98-33-16-80
- IPv4 Configuration:** DHCP is selected. Fields for IP Address, Subnet Mask, and Gateway are present but empty.
- Vap List:** A table with 7 columns: SSID, Encryption, Status, VAP MAC, Channel, and Radio Type. It lists two VAPs for HCIA-WLAN.

	SSID	Encryption	Status	VAP MAC	Channel	Radio Type
	HCIA-WLAN	NULL	Connected	00-E0-FC-EB-6A-80	1	802.11bgn
	HCIA-WLAN	NULL	Disconnected	00-E0-FC-EB-6A-C0	149	

Buttons: Connect, Disconnect, Refresh, Apply

```
STA>ping 10.0.1.1
```

```
Ping 10.0.1.1: 32 data bytes, Press Ctrl_C to break
```

```
From 10.0.1.1: bytes=32 seq=1 ttl=255 time=375 ms
```

```
From 10.0.1.1: bytes=32 seq=2 ttl=255 time=297 ms
```

```
From 10.0.1.1: bytes=32 seq=3 ttl=255 time=297 ms
```

```
From 10.0.1.1: bytes=32 seq=4 ttl=255 time=500 ms
```

```
From 10.0.1.1: bytes=32 seq=5 ttl=255 time=313 ms
```

```
--- 10.0.1.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
    round-trip min/avg/max = 297/356/500 ms
```

[AC]dis station all

Rf/WLAN: Radio ID/WLAN ID

Rx/Tx: link receive rate/link transmit rate(Mbps)

STA MAC address	AP ID SSID	Ap name	Rf/WLAN	Band	Type	Rx/Tx	RSSI	VLAN	IP a
--------------------	---------------	---------	---------	------	------	-------	------	------	------

5489-9833-1680	1	ap2	0/1	2.4G	-	-/-	-	101	192.
168.101.253	HCIA-WLAN								

Total: 1 2.4G: 1 5G: 0

Вывод

В ходе выполнения данной лабораторной работы мы сконфигурировали WLAN с помощью контроллера доступа (AC) и точек доступа AP, чтобы обеспечить доступ STA к WLAN.